Exercise

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Lemma 1. *Let $R \subset T$ be domains. In saying that T is algebraic over R, we mean that every element of T is algebraic over R; this is equivalent to saying that the quotient field of T is algebraic over the quotient field of R.*

Lemma 2.
- *Let R be a domain. Suppose that a is in a larger domain and is algebraic over R. Then $R[a]$ is algebraic.*
- *Let $R \subset T \subset S$ be domains. If S is algebraic over T and T is algebraic over R, then S is algebraic over R.*

Exercise 1. Let $R \subset T$ be domains with T finitely generated as a ring over R, and not algebraic over R. Prove that T is not a G-domain.

Proof. Let $T = R[t_1, \cdots, t_r, \cdots, t_n]$ where $t_1, \cdots, t_r$ are transcendental elements over R and $t_{r+1}, \ldots, t_n$ are algebraic elements over R, which are algebraic over $R[t_1, \cdots, t_r]$ all the more so. Write

$$R \subset R[t_1] \subset R[t_1, t_2] \subset \cdots \subset R[t_1, \cdots, t_r] \subset T.$$

Assume that T is a G-domain and then so is $R[t_1, \cdots, t_r]$. Without loss of generality, assume that t_r is transcendental over $R[t_1, \cdots, t_{r-1}]$. Hence we get a contradiction. □

Exercise 2. Let R be a domain with quotient field K, and u a non-zero element of R. Let J be the kernel of the homomorphism μ from $R[x]$ into K given by $x \mapsto u^{-1}$. Prove: $J = (ux - 1)$.

Proof. Since $\mu(ux - 1) = 0$, $(ux - 1) \subseteq \ker \mu = J$.

The reverse inclusion might not be as trivial as it seems. We use induction on the degree n of $f(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0 \in J$. When $n = 1$, $f = ax + b$; it follows from $a + bu = 0$ that $f = b - bux = -b(ux - 1) \in (ux - 1)$. Assume it holds for $n = r$. Write

$$f + a_0(ux - 1) = a_{r+1}x^{r+1} + \cdots + a_0 + a_0(ux - 1) = xg, \qquad \deg g = r.$$

Since $x \notin (ux - 1)$ and $f + a_0(ux - 1) \in J$ is prime, $g \in J$. However, by hypothesis, $g \in (ux - 1)$, hence f. □

Exercise 3. Let R be an integral domain having only a finite number of prime ideals. Prove that R is a G-domain.

Proof. Let $\mathfrak{p}_1, \ldots, \mathfrak{p}_r$ be non-zero prime ideals in R. Choose $0 \neq a_i \in \mathfrak{p}_i$, then $0 \neq a_1 \cdots a_r \in \bigcap_{i=1}^r \mathfrak{p}_i$ (since R is a domain). Theorem 2 applies and the proof is complete. □

Lemma 3. *Let $\mathfrak{a}_i$, $i = 1, \ldots, n$, be ideals in a commutative ring R. Suppose that $\mathfrak{p}$ is a prime ideal in R containing $\bigcap_i \mathfrak{a}_i$. Then for some $1 \leq k \leq n$, we have $\mathfrak{p} \supseteq \mathfrak{a}_k$.*

Proof. Assume that $\mathfrak{a}_k \nsubseteq \mathfrak{p}$ for each k. Therefore there exists $x_k \in \mathfrak{a}_k$ such that $x_k \notin \mathfrak{p}$. Then $x_1 \cdots x_r \in \bigcap_i \mathfrak{a}_i \subseteq \mathfrak{p}$, a contradiction. □

Corollary 1. *If $\bigcap_{i=1}^r \mathfrak{a}_i$ is a prime ideal $\mathfrak{p}$, then $\mathfrak{p} = \mathfrak{a}_k$ for some $1 \leq k \leq r$.*

Exercise 4. Let R be a Hilbert ring having only a finite number of maximal ideals. Prove that these are the only prime ideals in R.

Proof. By Corollary 11, a prime ideal is the intersection of maximal ideals containing it. Since R has a finite number of maximal ideals, by the corollary above, it follows that a prime ideal is equal to some maximal ideal. Therefore, these maximal ideals are the only prime ideals in R. □

Exercise 5. Let R be an integral domain with quotient field K. Assume that K is countably generated as a ring over R. Prove that R is a G-domain if and only if K cannot be expressed as the union of a properly ascending sequence of subrings containing R.

Proof. If R is a G-domain, then $K = R[u]$ for some $u \in K$. If K admits a form

$$K = \bigcup_{i=1}^{+\infty} R_i, \quad R_i \subsetneq R_{i+1},\ R_i \subsetneq K,\ \forall i,$$

then $u \in R_i$ for some i. It follows that $R[u] \subseteq R_i \subsetneq K$, a contradiction.

Conversely, assume that K cannot be expressed as the union of a properly ascending sequence of subrings containing R. Write $K = R[(a_j)_{j\in\mathbb{N}}]$. Clearly we have

$$K = \bigcup_{i=1}^{+\infty} R[(a_j)_{j\leq i}] = \bigcup R_j, \quad R_j \subseteq R_{j+1} \subseteq K.$$

By assumption, there exists finite j such that $R_j = K$. Hence K is finitely generated over R and then is a G-domain. □

Exercise 6. Let K be any field (not necessarily algebraically closed). Prove that any maximal ideal in $K[x_1, \cdots, x_n]$ can be generated by n elements.

Proof. Use induction on n; $K[x_1]$ is a PID so any maximal ideal can be generated by one element. Assume that the statement holds when $n = k$; when $n = k + 1$, let $\mathfrak{m}$ be a maximal ideal in $K[x_1, \cdots, x_{k+1}] = K[x_1, \cdots, x_k][x_{k+1}]$. Then $\mathfrak{m} \cap K[x_1, \cdots, x_k]$ is a G-ideal, hence a maximal ideal. By induction hypothesis, it follows that $\mathfrak{m} \cap K[x_1, \cdots, x_k]$ can be generated by k elements. By Theorem 13, $\mathfrak{m}$ can be generated by $(k + 1)$ elements. □

Exercise 7. Let R be a Hilbert ring such that every maximal ideal in R can be generated by k elements, k fixed. Prove that any maximal ideal in $R[x_1, \cdots, x_n]$ can be generated by $k + n$ elements.

Proof. Straightforward by induction. □

Exercise 8. Let R be a Hilbert ring in which every maximal ideal is finitely generated. Prove that the same is true for $R[x]$.

Proof. Straightforward by Theorem 13. □

Exercise 9. Prove that the following statements are equivalent:

(a) R is a Hilbert ring;

(b) Every radical ideal in R is an intersection of maximal ideals;

(c) Every prime ideal in R is an intersection of maximal ideals;

(d) Every G-ideal in R is an intersection of maximal ideals.

Proof.
- (a) implies (b): A radical ideal is equal to the intersection of G-ideals containing it; in a Hilbert ring, a G-ideal is a maximal ideal.
- (b) implies (c): A prime ideal is radical.
- (c) implies (d): A G-ideal is defined to be prime.
- (d) implies (a): $\mathfrak{p} = \bigcap \mathfrak{m}_i$. Since $R/\mathfrak{p}$ is a G-domain, there is a non-zero element contained in each non-zero prime ideal in $R/\mathfrak{p}$. Assume that for each i, $\mathfrak{p}$ is properly contained in $\mathfrak{m}_i$. Thus $\overline{\mathfrak{m}_i} \neq \{0\}$ is a non-zero prime ideal. However $\bigcap_i \overline{\mathfrak{m}_i} = \overline{\bigcap_i \mathfrak{m}_i} = \overline{\mathfrak{p}} = \{0\}$, a contradiction. Hence $\mathfrak{p}$ is equal to one of the $\mathfrak{m}_i$'s, hence a maximal ideal.

□

Exercise 10. Let R be any commutative ring. Let $\mathfrak{m}$ be a maximal ideal in $T = R[x_1, \cdots, x_n]$ satisfying $\mathfrak{m} \cap R = 0$. Prove that $\mathfrak{m}$ can be generated by $n+1$ elements.

Proof. Denote by v_i the image of x_i under the canonical homomorphism $T \twoheadrightarrow T/\mathfrak{m}$. As $\mathfrak{m} \cap R = 0$, $T/\mathfrak{m} \cong R[v_1, \cdots, v_n]$. Therefore R is a G-domain. Let K be the quotient field of R and it follows from R being a G-domain that $K = R[u^{-1}]$ for some $u \in R$. As $T/\mathfrak{m}$ is a field there exists $g \in T$ such that $ug \equiv 1 \pmod{\mathfrak{m}}$. Pass to the ring $T/(ug-1) \cong R[u^{-1}, x_1, \cdots, x_n] \cong K[x_1, \cdots, x_n]$. Therefore, by Exercise 6, $\mathfrak{m} \bmod (ug-1)$ can be generated by n more elements. Hence $\mathfrak{m}$ can be generated by $n+1$ elements. □

Exercise 11. Let K be algebraically closed and $R = K[x_1, \cdots, x_n]$. Let $f_1, \ldots, f_m \in R$. Then the equations

$$f_1 = 0,\ f_2 = 0,\ \ldots,\ f_m = 0$$

have a simultaneous solution if and only if there do not exist $g_1, \ldots, g_m \in R$ with

$$\sum g_i f_i = 1.$$

Proof. If $\sum g_i f_i = 1$ then clearly f_i have no simultaneous solutions. Conversely, if $(f_1, \cdots, f_m) \neq (1)$, there exists a maximal ideal $\mathfrak{m} \supseteq (f_1, \cdots, f_m)$. By the Nullstellensatz, $\mathfrak{m}$ admits a form $(x_1 - a_1, \ldots, x_n - a_n)$; i.e. for any $f \in \mathfrak{m}$, it follows that $f(a_1, \cdots, a_n) = 0$. Hence $f_1, \ldots, f_m$ have a simultaneous solution. □

Exercise 12. Let K be a field and write A for the vector space of n-tuples over K. Given an ideal $I \subset K[x_1, \cdots, x_n]$ we write $V(I)$ for the subset of A consisting of all points in A where all the polynomials in I vanish, and we call such a set a *variety*. Given a subset S of A, we write $J(S)$ for the ideal of polynomials vanishing on S. Assume that K is algebraically closed.

(1) Prove that $J(V(I))$ is the radical of I.

(2) Prove that the map $S \to J(S)$ sets up a one-to-one correspondence between all varieties and all radical ideals.

(3) Prove that under set-theoretic inclusion the set of radical ideals is a complete lattice and that the set of varieties likewise is a complete lattice. Prove further that the correspondence of part (2) is order-inverting.

(4) In the lattice of varieties of part (3), prove that the union of a finite number of varieties is just the set-theoretic union.

(5) Prove that in the correspondence of part (2) an ideal is prime if and only if the corresponding variety is *irreducible*, in the sense that it cannot be exhibited as the union of two properly smaller varieties.

Proof. • It follows from $f \in \sqrt{I} \iff f^n \in I$ that $\sqrt{I} \subseteq J(V(I))$. By the Nullstellensatz, the reverse inclusion is true also.

• Clear.

• Set $\mathcal{J} = \{\text{all radical ideals in } R\}$. For any subset $\mathcal{S}$ of $\mathcal{J}$, the ideal $\bigcap I_s$ is expected to be the $\bigwedge \mathcal{S}$. It is clear that the intersection of radical ideals is still a radical ideal. For any $J \in \mathcal{J}$, $J \subseteq I_s$, it follows that $J \subseteq \bigcap I_s$. Hence $\bigcap I_s = \bigwedge \mathcal{S}$. Define $\bigvee \mathcal{S}$ to be the radical ideal $\bigwedge\{J \in \mathcal{J} : I_s \subseteq J, \forall I_s \in \mathcal{S}\}$. By part (1), it is harmless to write a variety $V(I)$ where I is a radical ideal. It is clear that $\bigwedge\{V(I_s)\} = V(\bigvee \mathcal{S})$ and that $\bigvee\{V(I_s)\} = V(\bigwedge \mathcal{S})$. Thus the set of varieties is a complete lattice too. The last statement is straightforward.

• Suppose $p \notin V(I_1)$, $p \notin V(I_2)$, then there exist $f_1 \in I_1$, $f_1(p) \neq 0$ and $f_2 \in I_2$, $f_2(p) \neq 0$. Thus $f_1 f_2 \in I_1 \cap I_2$ with $f_1 f_2(p) \neq 0$; i.e. $p \notin V(I_1 \cap I_2)$, and hence $V(I_1 \cap I_2) \subseteq V(I_1) \cup V(I_2)$. The reverse inclusion is clear.

• Write $X = V(I)$ where I is a radical ideal. Suppose that X is irreducible and that $fg \in J(X) = I$. Let $Y = V(fg) = V(f) \cup V(g)$. Clearly $X \subseteq Y$ and it follows that $X = (V(f) \cap X) \cup (V(g) \cap X)$. Since X is irreducible, either $V(f) \supseteq X$ or $V(g) \supseteq X$, implying that either f or g is in I. Thus I is prime. The converse is clear.

□

Exercise 13. The homogeneous Nullstellensatz. K is a field, $R = K[x_1, \cdots, x_n]$. Call an ideal I in R *homogeneous* if along with a polynomial f, I contains the homogeneous constituents of f. Call a variety a *cone* if $(a_1, \cdots, a_n) \in V$ implies that $(ta_1, \cdots, ta_n) \in V$ for all $t \in K$.

(1) If I is homogeneous, prove that $V(I)$ is a cone.

(2) If V is a cone, and K is infinite, prove that $J(V)$ is homogeneous. Give an example to show that we need to assume that K is infinite.

(3) Thus or otherwise, prove that when K is algebraically closed, the radical of a homogeneous ideal is homogeneous. (This is actually true for any field.)

Proof. • Write $f = \sum_{i=0}^{N} f_i$ where f_i is a homogeneous polynomial of total degree i. Since I is homogeneous and $f \in I$, we have $f_i \in I$ for each i. By definition, for any $(a_1, \cdots, a_n) \in V(I)$, it follows that $f_i(a_1, \cdots, a_n) = 0$. As f_i is homogeneous, $f_i(ta_1, \cdots, ta_n) = 0$ for all $t \in K$.

• Example: Consider the finite field $\mathbb{Z}/3\mathbb{Z}$. Let R be $(\mathbb{Z}/3\mathbb{Z})[X]$. The principal ideal $I = (x^3 + 2x)$ is not homogeneous since it does not contain its homogeneous piece $2x$ while it vanishes on $\{0, 1, 2\}$ which is a cone. As $x^3 + 2x = x(x+1)(2+x)$, $J(\{0, 1, 2\}) = (x^3 + 2x)$.

- Suppose that $f = \sum_{i=0}^{N} f_i \in J(V)$ and $a \in V$. Denote $f_i(a)$ by a_i. It follows from $f(ta) = 0$ for all $t \in K$ that

$$a_N t^N + a_{N-1} t^{N-1} + \cdots + a_1 t + a_0 = 0, \qquad \forall t \in K.$$

  Hence,

$$a_N X^N + \cdots + a_0 \in K[X]$$

  has infinitely many solutions since K is infinite, which entails all a_i's being zero.

- Let I be a homogeneous ideal of $K[x_1, \cdots, x_n]$ where K is algebraically closed. $V(I)$ is a cone by part (1); by part (2), $J(V(I)) = \sqrt{I}$ is homogeneous.

□

Exercise 14. Prove: an element x in a ring R is nilpotent if and only if it is a zero-divisor on every R-module.

Proof. Let M be an R-module and $m \in M$. If x is nilpotent, then there exists n such that $x^n m = 0$ while $x^{n-1} m \neq 0$. Then $x(x^{n-1} m) = 0$ and x is a zero divisor. Conversely, suppose that x is a zero-divisor on each $R/\mathfrak{p}$. Then $x \in \bigcap \mathfrak{p}$ and x is nilpotent. □

Exercise 15. Let K be a field, L a field containing K, which is finitely generated as a ring over K. Prove that L is finite-dimensional over K.

Proof. Write $L = K[a_1, \cdots, a_n]$ where $a_i \in L$. By Theorem 7, it follows that $K[a_1, \cdots, a_{n-1}]$ is a G-domain and a_n is algebraic over it. Thus $K[a_1, \cdots, a_n]$ is finitely generated as a module over $K[a_1, \cdots, a_{n-1}]$. By induction on n, we have L finitely generated as a vector space over K. □

Exercise 16. Let $R \subset T$ be rings with R a Hilbert ring and T finitely generated as a ring over R. Prove that any maximal ideal in T contracts to a maximal ideal in R.

Proof. We have

$$R \xrightarrow{\iota} R[x_1, \cdots, x_r] \xrightarrow{\pi} R[x_1, \cdots, x_r]/I \cong T.$$

By Theorem 12 and Theorem 17, it immediately follows that any maximal ideal in T contracts to a maximal ideal in R. □

Exercise 17. Let $R_1 \subset R_2 \subset R_3$ be rings with R_3 finitely generated as a ring over R_1. Let I_3 be an ideal in R_3 and I_1, I_2 its contractions to R_1, R_2. Prove: if I_1, I_3 are maximal, so is I_2.

Proof. Clearly we have

$$R_1/I_1 \hookrightarrow R_2/I_2 \hookrightarrow R_3/I_3.$$

The exercise above applies to $R_1/I_1 \hookrightarrow R_3/I_3$ since R_3 is finitely generated as an R_1-algebra. Then we have R_3/I_3 is a finite field extension of R_1/I_1, which entails R_2/I_2 being algebraic over the field R_1/I_1. By Lemma 13, R_2/I_2 is a field and thus I_2 is maximal. □